1. Item 1

> Perform a simple calculation

The core of this problem is a straightforward arithmetic operation:
Given: Highest point height = 150 m, Diameter = 140 m,
Vertical distance from the highest point to the center M = Radius = 70 m.
Therefore, the height of M = 150 - 70 = 80 m.
The answer is obtained directly through simple calculation.

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2. Item 6

> Develop mathematical diagrams, graphs, constructions or computing artifacts and extracting mathematical information from them

The problem provides a diagram of the Ferris wheel. Students must extract key information from it:

- The highest point's height above the riverbed
- The labeled diameter
- The position of the center M
  and establish the geometric relationships from these.

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3. Item 7

> Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

Students need to process geometric information (diameter, height), perform numerical operations (subtraction, radius calculation), and understand the vertical geometric relationship between the highest point and the center.

4. Item 4

> Make mathematical diagrams, graphs, constructions or computing artifacts

Although a diagram is provided, students may need to **mentally or on scratch paper construct an auxiliary geometric model** to help establish the vertical relationship among "highest point – center – riverbed".

5. Item 12

> Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

The solution process implicitly involves two steps:

1. Calculate the radius: $140 \div 2 = 70$ m
2. Subtract the radius from the highest point: $150 - 70 = 80$ m
   This constitutes a simple but explicit multi-step reasoning process.

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6. Item 13

> Use an understanding of the context to guide or expedite the mathematical solving process, e.g. working to a context-appropriate level of accuracy

Students need to understand the real-world geometric meaning of "Ferris wheel above the riverbed," "highest point," and "center of the wheel," avoiding misconceptions such as assuming the center is at the riverbed or that the highest point equals the diameter plus the riverbed height.

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7. Item 10

> Make sense of, relate and use a variety of representations when interacting with a problem

The problem presents information in three forms: **picture + labeled diagram + text description**. Students must connect these representations to understand the relationships between "external diameter," "highest point," and "riverbed."

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8. Item 2

> Select an appropriate strategy from a list

Although the strategy for this problem is relatively direct (radius calculation + subtraction), students essentially need to choose the most straightforward method from several possible geometric or arithmetic approaches.

1. 第 1 条

> Perform a simple calculation

本题的核心是一步简单的算术运算：
已知最高点高度 = 150 m，直径 = 140 m，
则圆心 M 离最高点的垂直距离 = 半径 = 70 m。
因此 M 的高度 = 150 - 70 = 80 m。
直接依赖简单计算得出答案。

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2. 第 6 条

> Develop mathematical diagrams, graphs, constructions or computing artifacts and extracting mathematical information from them

题目给出了摩天轮的示意图，学生需要从图中提取关键信息：

- 最高点相对于河床的高度
- 直径标注
- 圆心 M 的位置
  并从中建立几何关系。

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3. 第 7 条

> Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

学生需要处理几何信息（直径、高度）、进行数值运算（减法、半径计算），并理解圆心与最高点之间的垂直几何关系。

4. 第 4 条

> Make mathematical diagrams, graphs, constructions or computing artifacts

虽然题目已提供图示，但学生在理解过程中可能需要**在脑中或草稿上构建辅助几何模型**，帮助建立“最高点 - 圆心 - 河床”的垂直关系。

5. 第 12 条

> Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

解题过程隐含两个步骤：

1. 计算半径：$140 \div 2 = 70$ m
2. 从最高点减去半径：$150 - 70 = 80$ m
   属于简单但明确的多步推理。

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6. 第 13 条

> Use an understanding of the context to guide or expedite the mathematical solving process, e.g. working to a context-appropriate level of accuracy

学生需要理解“摩天轮在河床上方”“最高点”“圆心”的实际几何意义，避免误以为圆心在河床上或最高点等于直径加河床高度。

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7. 第 10 条

> Make sense of, relate and use a variety of representations when interacting with a problem

题目以**图片 + 标注图 + 文字说明**三种方式呈现信息，学生需要将这些表征联系起来，理解“外部直径”“最高点”“河床”之间的关系。

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8. 第 2 条

> Select an appropriate strategy from a list

虽然本题策略较直接（半径计算 + 减法），但本质上学生需在多个可能的几何或算术方法中选择最直接的一种。